

CHITRACLOUD Web Application Using AWS Services

Abstract

ChitraCloud is a cloud-based image editing web application developed to provide a single platform for uploading, editing, converting, and storing images. In traditional systems, users face issues such as limited storage, lack of accessibility, and the need to use multiple tools for tasks like PDF to image conversion and editing. This project aims to solve these problems by integrating all functionalities into one system.

The application is built using Amazon Web Services (AWS), which provides scalable and secure cloud services. Key services used in this project include Amazon S3 for image storage, Amazon DynamoDB for metadata management, Amazon Cognito for user authentication, and AWS IAM for secure access control.

The purpose of deploying this application on the cloud is to ensure high availability, scalability, and secure data handling while allowing users to access the system from anywhere. The final outcome of the project is a user-friendly, efficient, and secure cloud-based image editing platform that simplifies image processing tasks and improves user experience.

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Chapter 1: Introduction

1. Introduction

Cloud Computing is a technology that allows users to store, manage, and process data over the internet instead of using local computers or servers. It provides on-demand access to computing resources such as storage, databases, and applications. Platforms like Amazon Web Services offer reliable and scalable cloud services.

Cloud computing is important because it reduces the need for physical hardware and allows users to access their data anytime and from anywhere. It improves flexibility, reduces cost, and provides better performance compared to traditional systems.

Traditional systems depend on local storage and installed software, which can lead to problems such as limited storage, data loss, and lack of accessibility. In contrast, cloud platforms provide secure storage, automatic backup, and scalability.

The objective of this project is to develop a cloud-based image editing application, ChitraCloud, that allows users to upload, edit, convert, and store images with secure and real-time processing.

1.1 Problem Statement

In today's digital world, users frequently need to edit and manage images. However, traditional image editing systems have several limitations. These systems require software installation, depend on local storage, and are not easily accessible from different devices.

To develop a cloud-based image editing application for uploading, editing, converting, and storing images.

The main problems in existing systems are:

- Lack of accessibility across devices
- Risk of data loss due to local storage
- No integrated platform for PDF to image conversion and editing
- Time-consuming process due to use of multiple tools

This project solves these problems by providing a single cloud-based platform where users can upload, convert (PDF to image), edit, and store images.

Cloud computing is suitable for this solution because it provides scalable, secure, and efficient services. In this project, services like Amazon S3 for image storage, Amazon DynamoDB for metadata storage, Amazon Cognito for user authentication, and AWS IAM for secure access control are used.

1.2 Objectives of the Project

The main objectives of this project are:

- To develop a cloud-based image editing web application
- To deploy and manage data using cloud infrastructure
- To use multiple cloud services such as Amazon S3, Amazon DynamoDB, and Amazon Cognito
- To implement secure image storage and user authentication
- To provide real-time image editing and PDF to image conversion
- To ensure scalability and availability using cloud services
- To improve user experience by providing all features in one platform
- To understand cloud security and resource management

Chapter 2: Cloud Platform Overview

The cloud platform used in this project is Amazon Web Services (AWS). AWS is one of the most popular and widely used cloud service providers. It offers a wide range of services for storage, computing, databases, and security.

AWS is selected for this project because it provides high scalability, reliability, and security. It also has easy integration with web applications and supports multiple services like Amazon S3, Amazon DynamoDB, and Amazon Cognito. Additionally, AWS provides a free tier, which is useful for students and small projects.

2.1 Cloud Deployment Model

This project uses the Public Cloud deployment model. In the public cloud, services are provided over the internet and shared among multiple users. AWS manages the infrastructure, and users can access services anytime from anywhere. This model is cost-effective, scalable, and easy to use.

2.2 Cloud Service Models Used

1. Infrastructure as a Service (IaaS)

IaaS provides basic computing resources like servers, storage, and networking. In this project, AWS infrastructure supports storage and backend services. It allows flexible resource usage without managing physical hardware.

2. Platform as a Service (PaaS)

PaaS provides a platform to develop, run, and manage applications without handling the underlying infrastructure. In this project, AWS services help in building and deploying the application easily.

3. Software as a Service (SaaS)

SaaS provides software applications over the internet. The ChitraCloud application itself acts as a SaaS product, where users can access image editing features through a web browser without installing any software.

Chapter 3: Services Used

In this project, multiple cloud services from Amazon Web Services are used to build a scalable and secure image editing application.

3.1 Amazon S3 (Simple Storage Service)

- **Service Category:** Storage

- **Purpose in the Project:**

Amazon S3 is used to store uploaded images and edited images securely in the cloud. It allows users to access their images anytime from anywhere.

- **Why this Service was Chosen:**

S3 provides highly scalable, durable, and secure storage. It is cost-effective and easy to integrate with web applications.

3.2 Amazon Cognito

- **Service Category:** Authentication / Security

- **Purpose in the Project:**

Amazon Cognito is used for user authentication. It allows users to sign up, log in, and securely access their personal data and images.

- **Why this Service was Chosen:**

Cognito provides secure and easy authentication with features like user management and access control. It reduces the need to build authentication systems from scratch.

3.3 AWS IAM (Identity and Access Management)

- **Service Category:** Security

- **Purpose in the Project:**

AWS IAM is used to control access to AWS services. It manages permissions and ensures that only authorized users and services can access resources like S3 and DynamoDB.

- **Why this Service was Chosen:**

IAM provides strong security by allowing role-based access control. It helps in protecting data and managing permissions efficiently.

3.4 Amazon DynamoDB

- **Service Category:** Database (NoSQL)

- **Purpose in the Project:**

Amazon DynamoDB is used to store metadata of images such as file name, size, upload time, and user details.

- **Why this Service was Chosen:**

DynamoDB is a fast, scalable, and fully managed NoSQL database. It provides high performance and is suitable for real-time applications.

Chapter 4: System Architecture

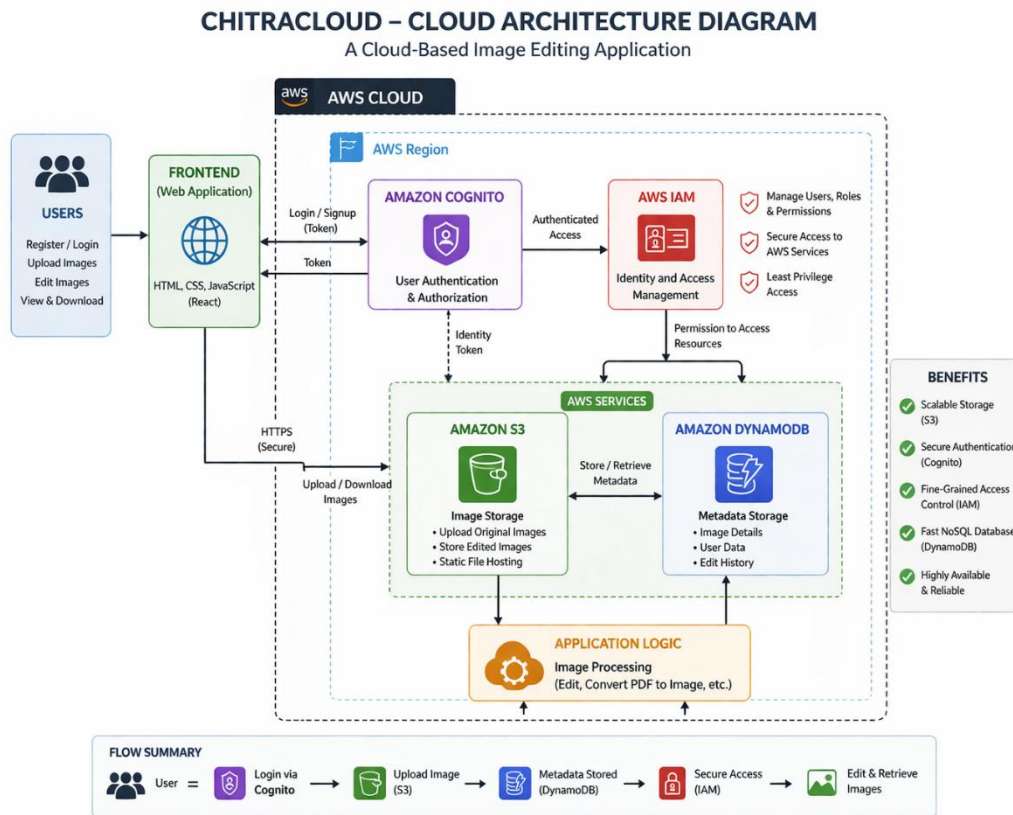


Fig. System Architecture

Description

The above diagram represents the cloud architecture of the ChitraCloud application. It shows how users interact with the system through a web-based frontend built using React.

When a user logs in or registers, authentication is handled by Amazon Cognito, which generates a secure token. This token is used to provide authenticated access to cloud resources.

The application uses Amazon S3 to store original and edited images. It ensures scalable and secure storage of user data. Metadata related to images, such as file details and user information, is stored in Amazon DynamoDB.

Security and access control are managed using AWS IAM, which ensures that only authorized users can access the resources. All communication between the frontend and cloud services is done over secure HTTPS.

The application logic handles image processing operations such as editing and PDF-to-image conversion. The diagram also shows the complete flow of data from user request to cloud services and back, ensuring efficient and secure operation of the system.

Chapter 5: Implementation Details

5.1 Technology Stack

- Frontend: React.js, HTML, CSS, JavaScript
- Backend: Node.js, Express.js
- Database: Amazon DynamoDB
- Cloud Platform & Services: Amazon Web Services (S3, Cognito, IAM, DynamoDB)

5.2 Implementation Steps

1. Cloud Account Setup

An AWS account was created using Amazon Web Services. Free tier services were enabled for cost-effective usage.

2. Resource Creation

The following resources were created:

- Amazon S3 bucket for storing images
- Amazon DynamoDB table for metadata
- Amazon Cognito user pool for authentication
- AWS IAM roles and policies for access control

3. Configuration Steps

- Configured S3 bucket permissions
- Set up DynamoDB table schema
- Configured Cognito user authentication
- Assigned IAM roles with required permissions

4. Code Deployment

- Frontend developed using React.js
- Backend developed using Node.js and Express
- Connected frontend with AWS services using APIs

5. Database Connection

- DynamoDB is connected using AWS SDK
- Metadata such as image name, size, and user details are stored and retrieved

5.3 Screenshots

The following screenshots show the implementation of the system:

- Amazon S3 Bucket (Image Storage)

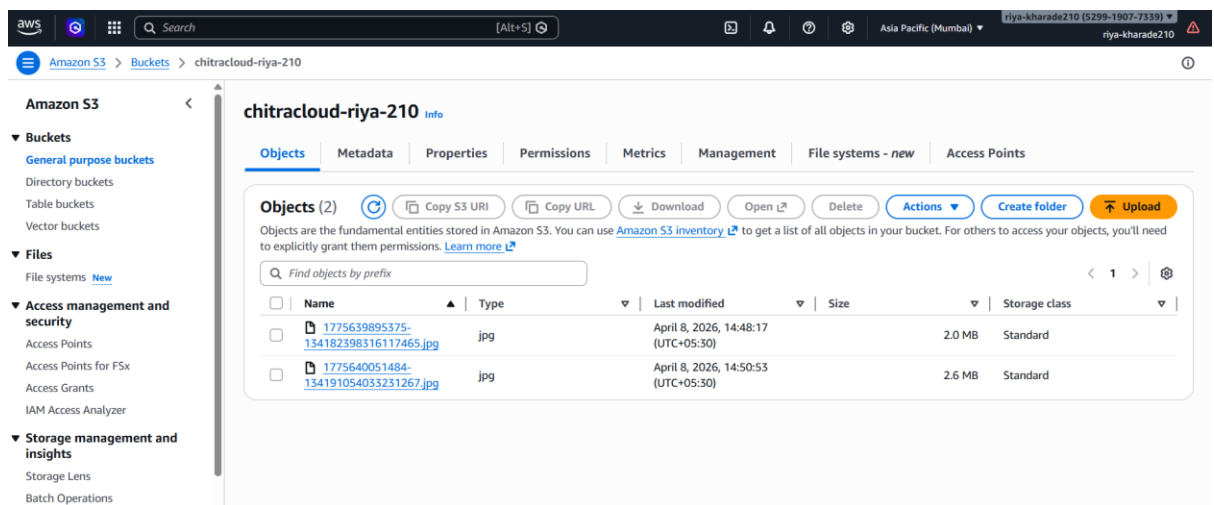


Fig 5.3.1. Amazon S3 Bucket

- Amazon DynamoDB Table

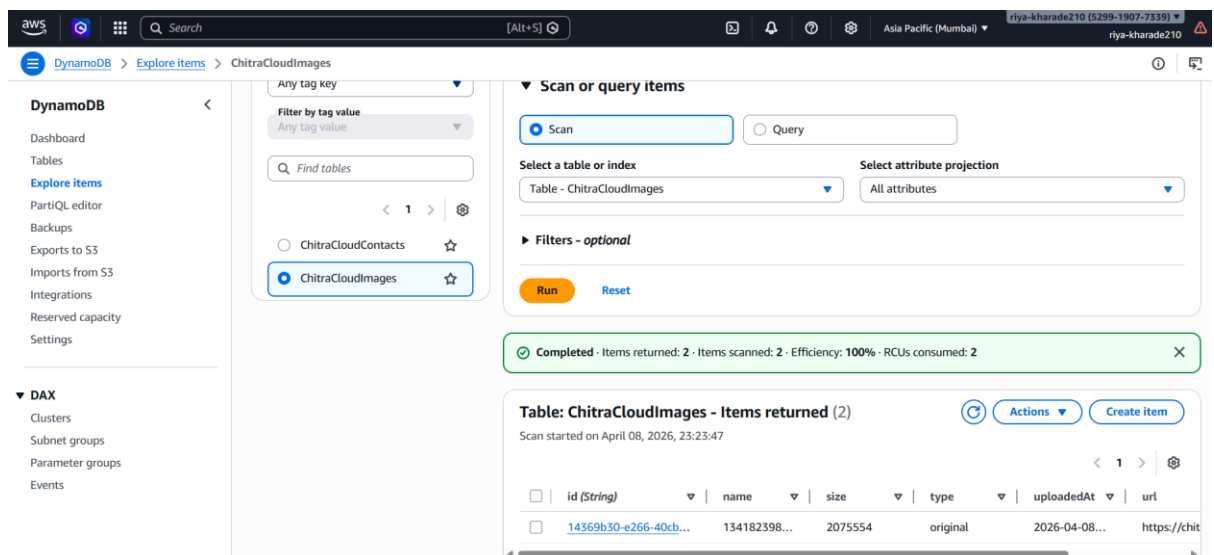


Fig 5.3.2. Amazon DynamoDB Table

- Amazon Cognito User Pool

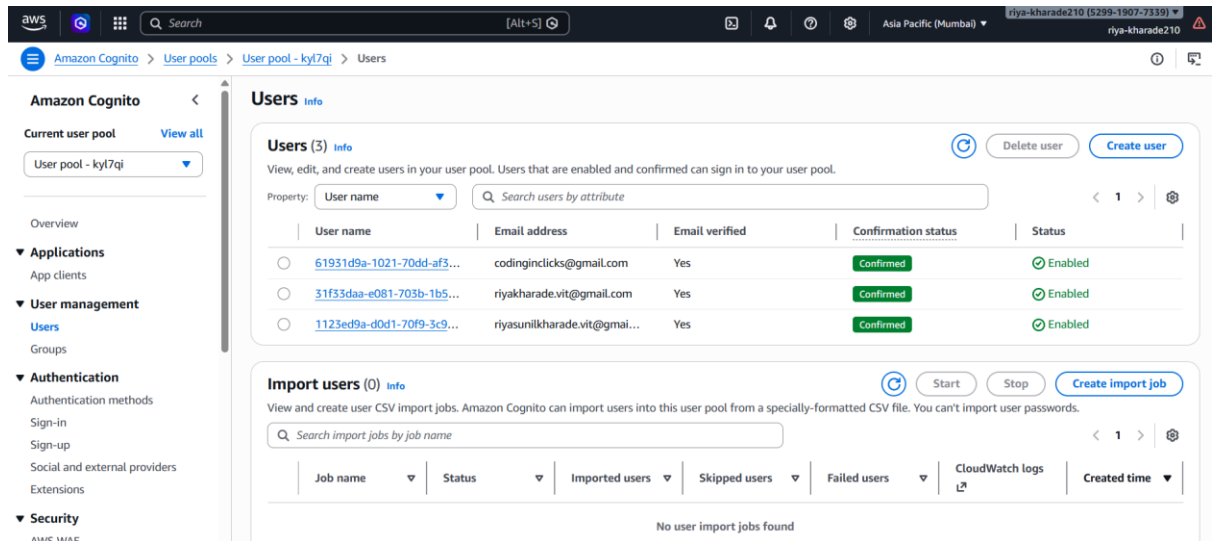


Fig 5.3.3. Amazon Cognito User Pool

- Amazon IAM

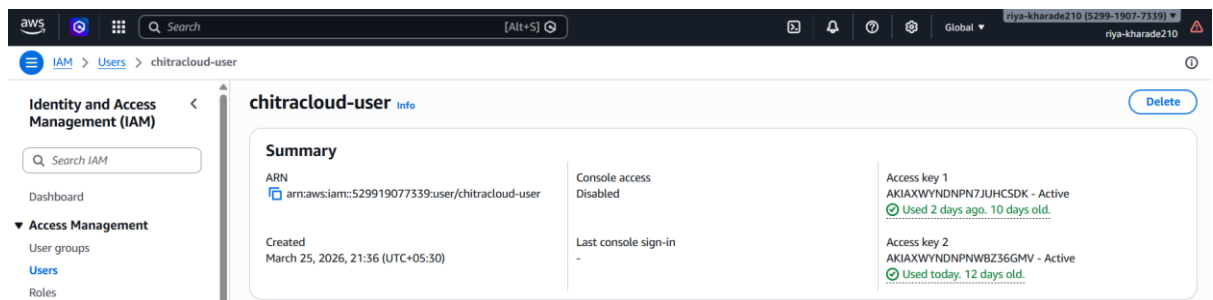


Fig 5.3.4. Amazon IAM

- Application Output (User Interface)

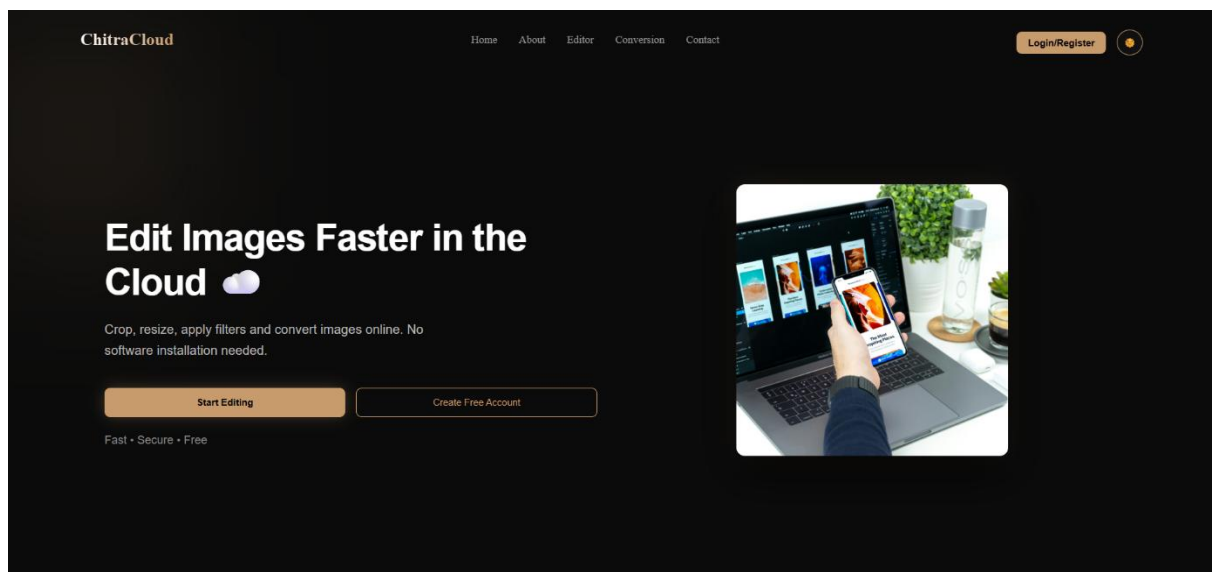


Fig 5.3.5. Home Page

5.4 Security Implementation

- **IAM Roles / Users:** AWS IAM is used to create roles and assign permissions to services.
- **Authentication & Authorization:** Amazon Cognito is used for secure login and user management.
- **Security (Access Control):** Access to resources is restricted using IAM policies and secure tokens.
- **Data Security:** Images are stored securely in Amazon S3, and communication is done using HTTPS.

5.5 Cost Analysis

- **Pricing Model Used:** AWS follows a Pay-as-you-go model, where users pay only for the resources they use.
- **Approximate Cost Estimation:** The project mainly uses free tier services, so the cost is minimal or zero.
- **Steps Taken to Minimize Cost:**
 - Used AWS Free Tier services
 - Avoided unnecessary resource usage
 - Optimized storage and requests
- **Free Tier Usage:** Services like S3, DynamoDB, and Cognito were used under free tier limits.
- **Resource Shutdown Strategy:** Unused resources are stopped or deleted to prevent extra charges.

Chapter 6: Testing & Results

6.1 Testing & Results

The developed system was tested to ensure that all functionalities are working correctly. Different types of testing were performed to verify the performance and reliability of the application.

Functional Testing

Functional testing was carried out to check whether all features of the application are working as expected. The following functionalities were tested:

- User registration and login using Amazon Cognito
- Image upload and storage in Amazon S3
- Image editing features such as filters, rotate, and adjustments
- PDF to image conversion
- Display of image metadata from Amazon DynamoDB
- Image download and deletion

All functionalities were tested successfully.

Performance Observation

The application showed good performance during testing. Image upload, editing, and retrieval operations were completed quickly. Cloud services ensured fast response time and smooth user experience.

Successful Deployment Confirmation

The application was successfully deployed and connected with AWS cloud services. All components such as frontend, backend, and cloud services were integrated properly and worked without errors.

Screenshots of Working Application

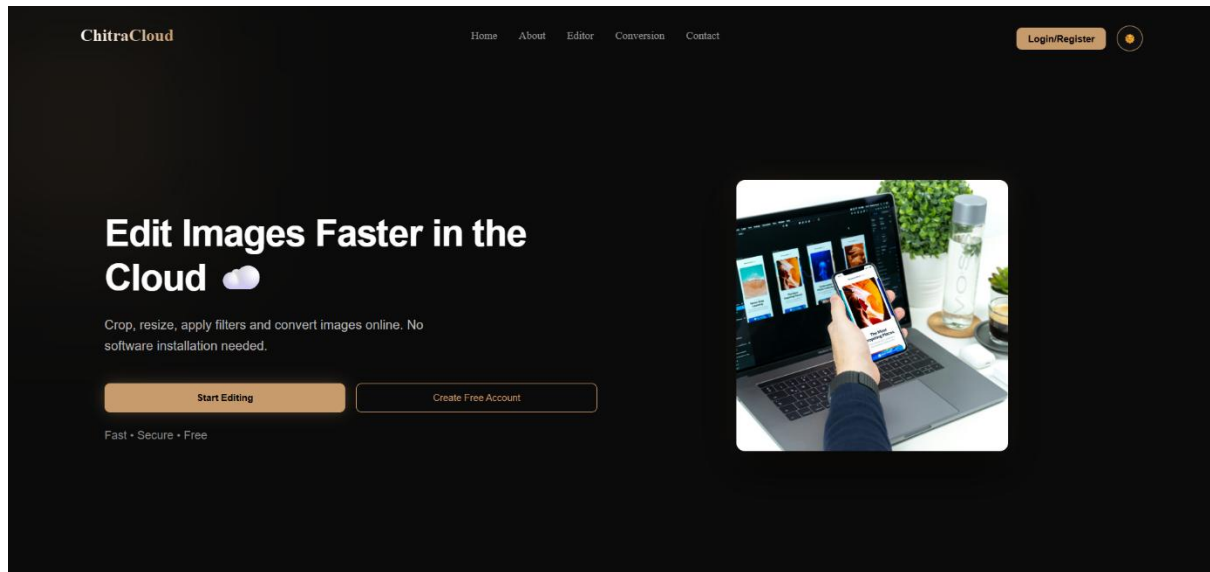


Fig 6.1.1 Home Page

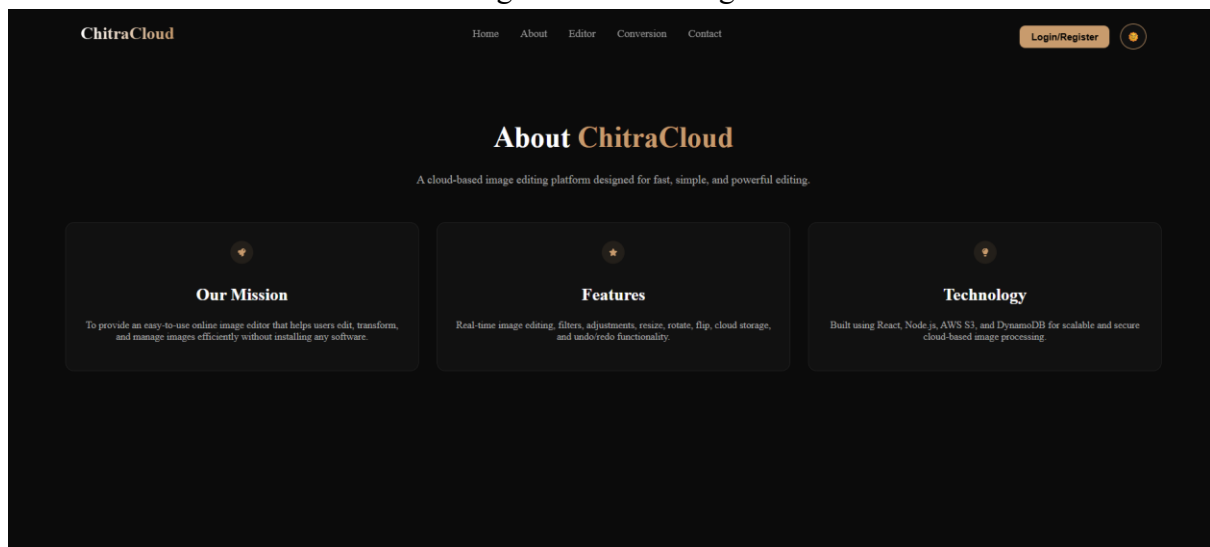


Fig 6.1.2 About ChitraCloud Page

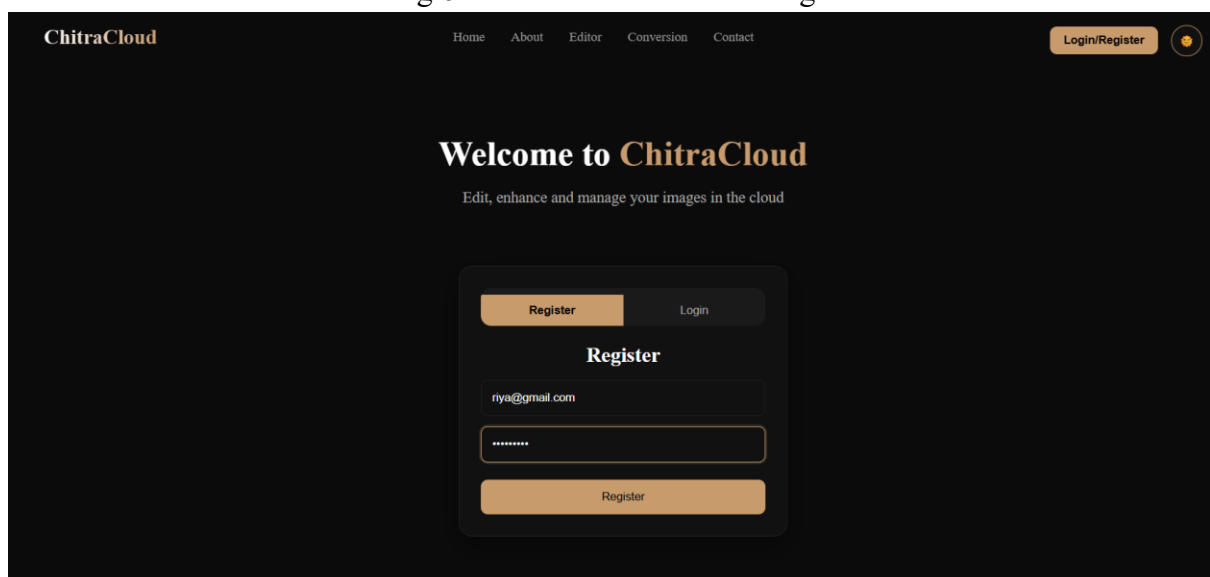
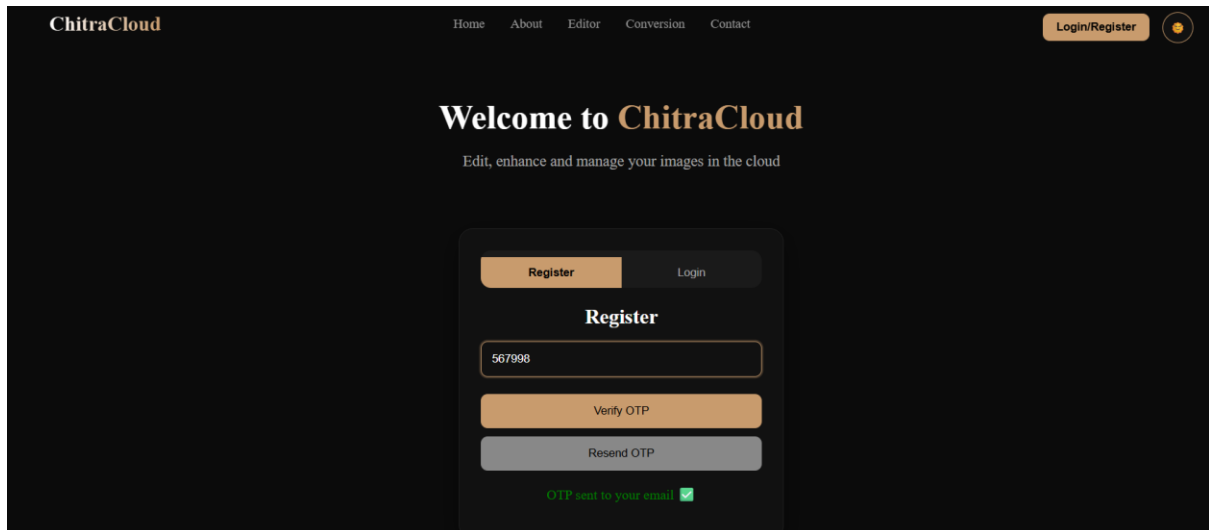
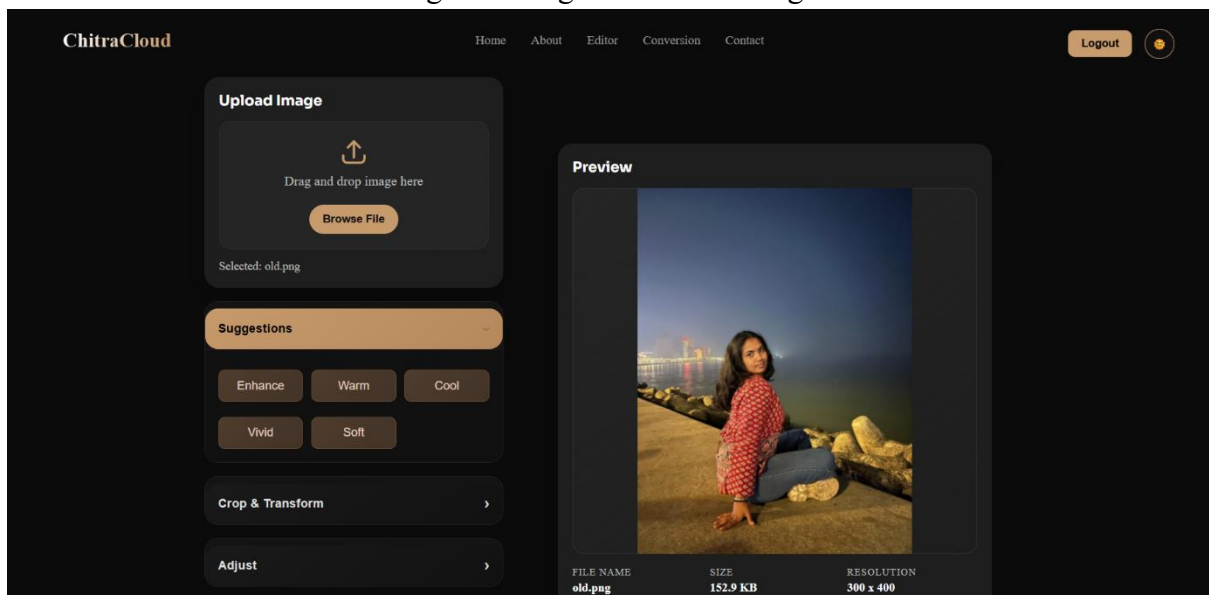


Fig 6.1.3. Registration Page



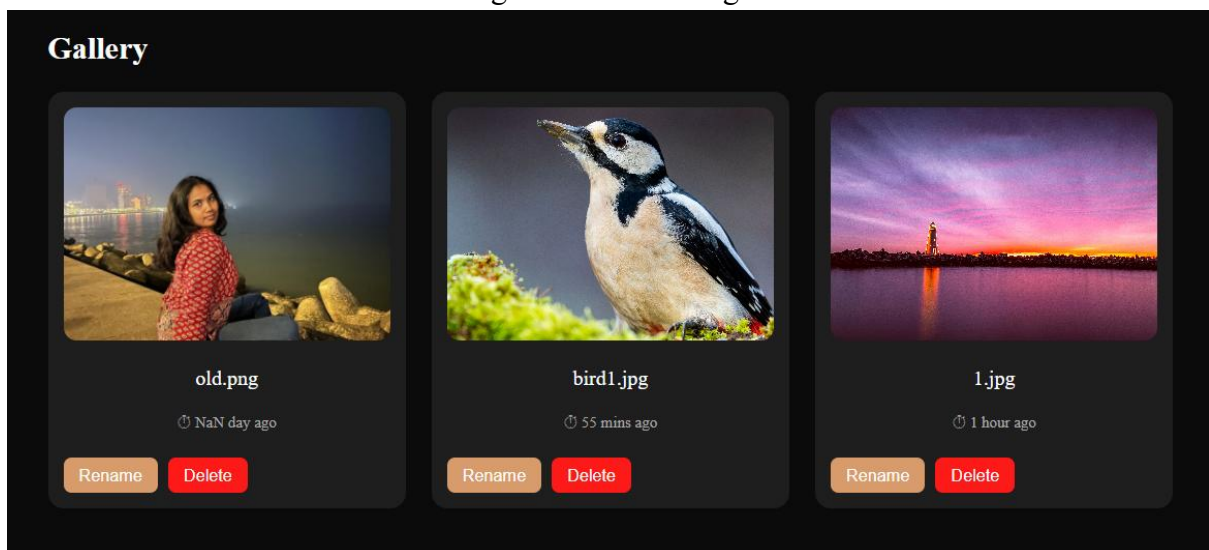
The image shows the registration OTP page of ChitraCloud. The page has a dark background with a light blue header containing the ChitraCloud logo and navigation links: Home, About, Editor, Conversion, and Contact. A 'Login/Register' button is in the top right corner. The main heading is 'Welcome to ChitraCloud' with the tagline 'Edit, enhance and manage your images in the cloud'. Below this is a registration form with a 'Register' button and a 'Login' button. The form contains a text input field with the value '567998', a 'Verify OTP' button, and a 'Resend OTP' button. A green message at the bottom says 'OTP sent to your email' with a checkmark icon.

Fig 6.1.4 Registration OTP Page



The image shows the editor page of ChitraCloud. The page has a dark background with a light blue header containing the ChitraCloud logo and navigation links: Home, About, Editor, Conversion, and Contact. A 'Logout' button is in the top right corner. The main content area is divided into two sections: 'Upload Image' and 'Preview'. The 'Upload Image' section has a 'Drag and drop image here' area with an upload icon and a 'Browse File' button. Below this, it says 'Selected: old.png'. The 'Preview' section shows a preview of the image 'old.png' with a file name, size (152.9 KB), and resolution (300 x 400). Below the preview, there are buttons for 'Enhance', 'Warm', 'Cool', 'Vivid', and 'Soft'. There are also buttons for 'Crop & Transform' and 'Adjust'.

Fig 6.1.5. Editor Page



The image shows the gallery page of ChitraCloud. The page has a dark background with a light blue header containing the ChitraCloud logo and navigation links: Home, About, Editor, Conversion, and Contact. A 'Logout' button is in the top right corner. The main heading is 'Gallery'. Below this, there are three image cards. Each card shows an image, a file name, a timestamp, and buttons for 'Rename' and 'Delete'. The first card shows a woman sitting on a rock, with file name 'old.png' and timestamp 'NaN day ago'. The second card shows a bird, with file name 'bird1.jpg' and timestamp '55 mins ago'. The third card shows a sunset, with file name '1.jpg' and timestamp '1 hour ago'.

Fig 6.1.6 Gallery Page

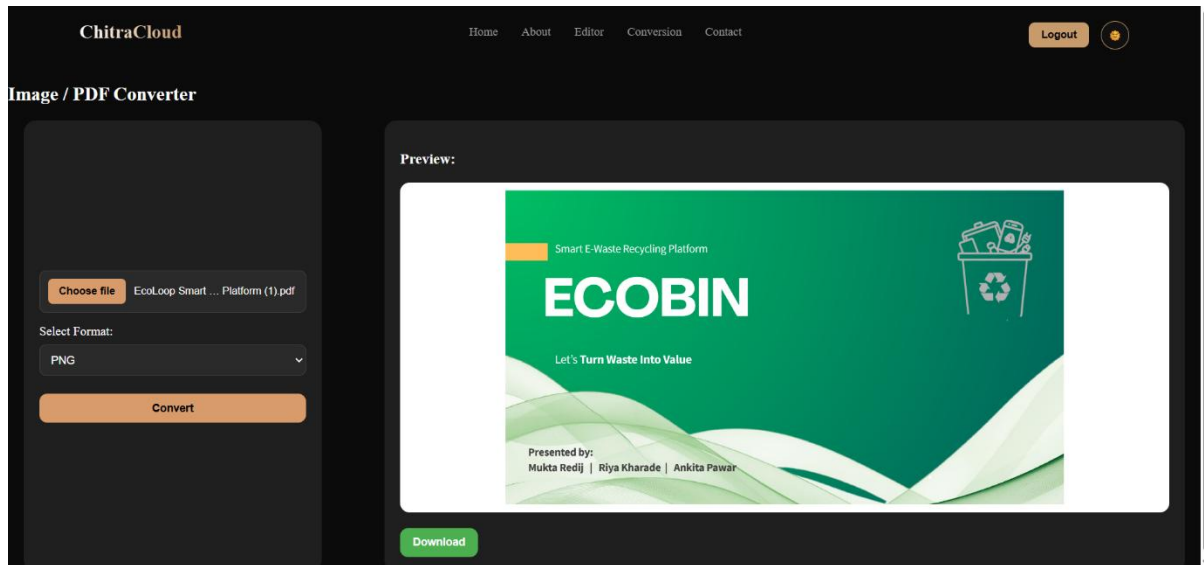


Fig 6.1.7 File Conversion Page

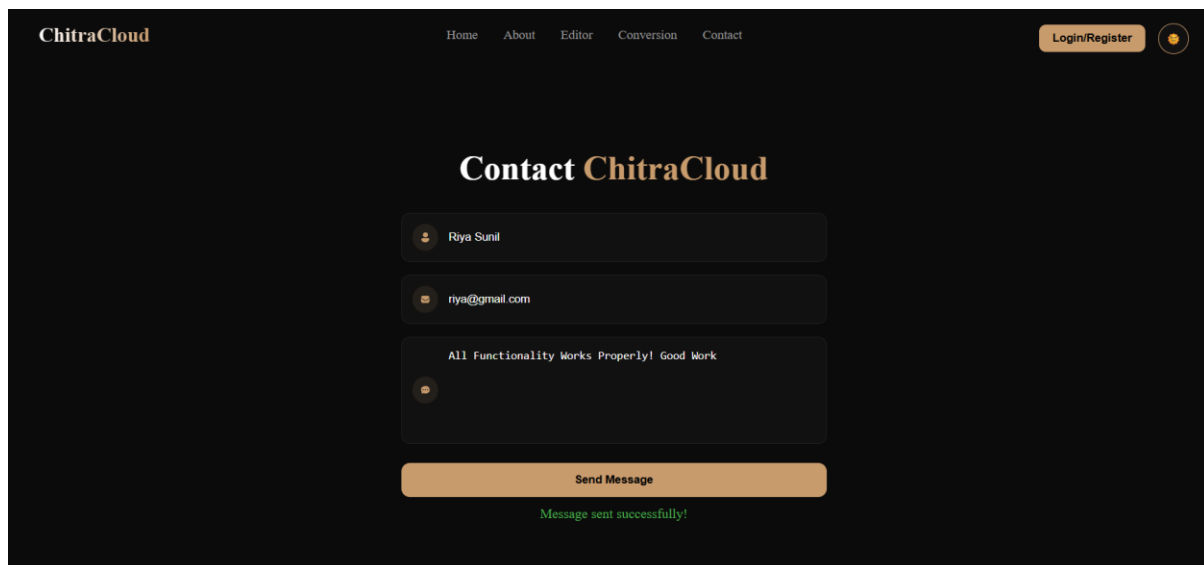


Fig 6.1.8. Contact Page

6.2 Advantages of Cloud-Based Solution

- Easy access from anywhere using internet
- Scalable storage using Amazon S3
- Secure authentication using Amazon Cognito
- High performance and reliability
- Integrated platform for image editing and PDF conversion

6.3 Limitations

- Requires internet connection to access the application
- Limited features compared to professional editing software
- Dependency on cloud services
- Possible cost if usage exceeds free tier

Chapter 7: Conclusion and Future Scope

7. Conclusion

The project **ChitraCloud** was successfully developed as a cloud-based image editing web application. The system allows users to upload, edit, convert (PDF to image), and store images efficiently using cloud technology.

In this project, various cloud services such as Amazon S3 for storage, Amazon DynamoDB for metadata, Amazon Cognito for authentication, and AWS IAM for security were used.

Through this project, we learned how to integrate cloud services into a real-world application, manage secure data storage, and implement authentication and access control. The project achieved its objective of providing a scalable, secure, and user-friendly image editing platform.

7.1 Future Scope

The project can be further improved by adding more advanced features such as:

- AI-based image enhancement and filters
- Multi-user collaboration features
- Integration with more cloud services
- Advanced editing tools like background removal

References

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